Advantages of Quantum Fourier Transform (QFT)

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Old slides from 2005, slightly revised

Introduction to Quantum Fourier Transform (QFT)

- QFT is the quantum analogue of the Discrete Fourier Transform (DFT).
- It plays a crucial role in quantum algorithms like Shor's Algorithm and Quantum Phase Estimation.
- QFT provides an **exponential speedup** over classical Fourier Transform methods.

Mathematical Definition of QFT

QFT on an n-qubit register:

$$\mathrm{QFT}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i x k / 2^n} |k\rangle$$

where x is the computational basis state.

Advantage 1: Exponential Speedup

- Classical Discrete Fourier Transform (DFT) requires $O(N^2)$ operations.
- The Fast Fourier Transform (FFT) improves this to $O(N \log N)$.
- QFT reduces complexity to **$O((\log N)^2)$** using quantum gates.

Advantage 2: Quantum Parallelism

- QFT acts on a **superposition** of states, processing all inputs simultaneously.
- This is crucial in algorithms like:
  - Shor's Algorithm for factoring large numbers.
  - Quantum Phase Estimation (QPE) for eigenvalue extraction.

Advantage 3: Compact Circuit Implementation

- QFT requires only **Hadamard gates and controlled-phase gates**.
- A 3-qubit QFT circuit uses only $O(n^2)$ gates.
- This makes QFT highly efficient for **quantum hardware**.

Advantage 4: Applications in Quantum Computing

- **Shor's Algorithm:** Uses QFT to find periodicity in modular exponentiation.
- **Quantum Phase Estimation (QPE):** Extracts eigenvalues of unitary matrices.
- **Quantum Signal Processing:** Enables spectral analysis on quantum data.

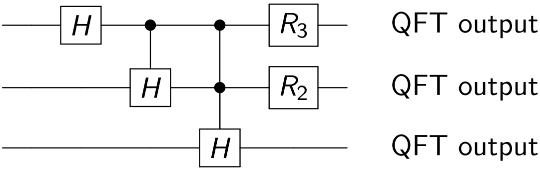
Advantage 5: Reduced Memory Complexity

- Classical DFT requires $O(N)$ space.
- QFT stores Fourier-transformed coefficients **implicitly** in qubit amplitudes.
- Only **$O(n)$ qubits** are needed for a size $N = 2^n$ transformation.

Advantage 6: Quantum Signal Processing

- QFT enables applications such as:
    - Quantum spectral analysis.
    - Quantum image processing.
    - Quantum filtering and denoising.
- These can enhance AI, cryptography, and data processing.

Quantum Fourier Transform Circuit



Conclusion

- Quantum Fourier Transform provides **exponential speedup**.
- It enables key quantum algorithms like **Shor's Algorithm** and **Quantum Phase Estimation**.
- QFT is more efficient in **memory usage and circuit complexity** than classical methods.
- Future quantum applications in **signal processing, AI, and cryptography** will heavily rely on QFT.